THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 5–8: **Life Processes, Control and Coordination, How do Organisms Reproduce?, Heredity** | Class 10 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** excretion	Kidneys filter blood and form urine.
2	**(d)** dendrite	Dendrite tip receives the signal.
3	**(b)** at a definite plane related to the whip-like flagellum	Leishmania has a definite orientation of division; Amoeba divides in any plane.
4	**(a)** Gregor Johann Mendel	Mendel worked on garden pea plants.
5	**(a)** CO_2, water and energy	Glucose is fully oxidised.
6	**(c)** phototropism	Positive phototropism.
7	**(a)** pollen grain	Pollen grains carry the male gametes.
8	**(a)** all round yellow	Round and yellow are dominant.
9	**(a)** nephron	Each kidney has about a million nephrons.
10	**(a)** adrenaline	Adrenal glands.
11	**(a)** testes	Testes are the male sex organs.
12	**(d)** the type of sperm that fertilises the egg	Father's sperm may carry X or Y.
13	**(d)** transport of water	Xylem carries water and minerals.
14	**(d)** synapse	Chemical signal crosses it.
15	**(c)** Plasmodium	Plasmodium divides into many daughter cells at once.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Autotrophs make their own food by photosynthesis (1); Heterotrophs depend on other organisms for food (1).	1+1
17	Reflex is automatic and involuntary; controlled by spinal cord (1); Walking is voluntary and controlled by the brain (1).	1+1
18	Variations help some individuals survive when the environment changes (1); This ensures the survival of the species over generations (1).	1+1
19	Heredity: transmission of characters from parents to offspring (1); Variation: differences among individuals of a species (1).	1+1
20	Long and coiled, with villi (1); Villi increase surface area and are rich in blood vessels (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Energy is released (1); $C_6H_{12}O_6 + 6\,O_2 \longrightarrow 6\,CO_2 + 6\,H_2O + \text{energy}$ (1); Oxidation of glucose (1).	1+1+1
22	Receptor in the skin detects heat (1); Sensory neuron to spinal cord; relay neuron; motor neuron (1); Effector muscle withdraws the hand (1).	1+1+1
23	A bud develops as an outgrowth due to repeated cell division (1); The bud grows into a tiny individual with its own mouth and tentacles (1); It detaches when mature and becomes an independent Hydra (1).	1+1+1
24	F1: genotype Tt, phenotype tall (1); F2 phenotypic ratio 3 tall : 1 dwarf (1); F2 genotypic ratio 1 TT : 2 Tt : 1 tt (1).	1+1+1
	OR: Filtration in glomerulus (1); Selective reabsorption of glucose, amino acids, water (1); Remaining waste forms urine (1).	1+1+1
25	Insulin: pancreas, sugar level (1); Thyroxine: thyroid, metabolism (1); Adrenaline: adrenal, emergency (1).	1+1+1
	OR: Anther produces pollen grains (1); Stigma receives pollen (1); Ovary contains ovules and becomes fruit (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Chlorophyll absorbs light energy (1); Water splits into hydrogen and oxygen (1); CO_2 is reduced to carbohydrates (1); Desert plants take CO_2 at night and store it as an acid (1); Use it during the day for photosynthesis (1).	1+1+1+1+1

	OR: Dendrites receive information (1); Cell body (1); Axon carries impulse away (1); Nerve endings release chemical (1); Synapse transmits to next neuron (1).	1+1+1+1+1
27	Fission (binary and multiple): Amoeba, Plasmodium (1); Fragmentation: Spirogyra (1); Regeneration: Planaria (1); Budding: Hydra, yeast (1); Spore formation: Rhizopus (1).	1+1+1+1+1
	OR: Parents: TT x tt (1); F1: all Tt, tall (1); F1 self-pollinated: gametes T and t (1); F2: TT, Tt, Tt, tt (1); Phenotypic ratio 3 : 1 and genotypic ratio 1 : 2 : 1 (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 5–8: **Life Processes, Control and Coordination, How do Organisms Reproduce?, Heredity** | Class 10 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** it has many contrasting characters and is easy to grow	Pea has clear contrasting traits and short life cycle.
2	**(a)** lactic acid	Lactic acid causes cramps.
3	**(a)** gravity	Positive geotropism.
4	**(d)** ovary	After fertilisation the ovary ripens into fruit.
5	**(b)** 9 : 3 : 3 : 1	Four phenotypes appear.
6	**(a)** urinary bladder	Bladder.
7	**(b)** pituitary	It controls other glands.
8	**(d)** sperm formation needs a temperature lower than body temperature	A lower temperature is essential for sperm production.
9	**(a)** XX	Both are X.
10	**(b)** carbon dioxide and water	Photosynthesis needs all of these.
11	**(d)** insulin	Insulin is an animal hormone.
12	**(d)** budding	A bud grows out as an outgrowth and detaches.
13	**(a)** heredity	Definition of heredity.
14	**(d)** ethanol and carbon dioxide	Fermentation.
15	**(b)** thigmotropism	Response to touch.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Reflex is controlled by the spinal cord (1); The brain is informed afterwards and does not control the action (1).	1+1
17	New plants are obtained from vegetative parts such as root, stem or leaf (1); Rose, Bryophyllum or sugarcane (1).	1+1
18	A cross involving one pair of contrasting characters (1); F2 ratio is 3 : 1 (1).	1+1
19	Xylem transports water and minerals (1); Phloem transports food (1).	1+1
20	Iodine is needed to make thyroxine (1); Prevents goitre (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Eggs always carry an X chromosome (1); Sperm can carry X or Y (1); The sperm that fertilises the egg decides the sex (1).	1+1+1
22	Aerobic: CO_2 and water, more energy (1); Anaerobic: ethanol or lactic acid, less energy (1); Aerobic in mitochondria (1).	1+1+1
23	Cerebrum: thinking (1); Cerebellum: balance (1); Medulla: involuntary actions (1).	1+1+1
24	Boys: growth of facial hair, voice change (1); Girls: breast development, onset of menstruation (1); Both: hair under the arm and genital area, sexual maturity (1).	1+1+1
	OR: DNA carries genetic information (1); Segments called genes control traits (1); DNA is passed from parents to offspring through gametes (1).	1+1+1
25	Water enters guard cells: they swell and the pore opens (1); When water leaves they shrink and the pore closes (1); This controls gas exchange and water loss (1).	1+1+1
	OR: Quick response to danger (1); Saves time as brain is not involved (1); Example: knee jerk or touching a hot plate (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks

26	Kidneys, ureters, bladder, urethra (1); Filtration in glomerulus (1); Reabsorption in tubules (1); Urine flows to bladder through ureter (1); Released through urethra (1).	1+1+1+1+1
	OR: Light causes auxin to shift to the shaded side (1); Cells on that side elongate more (1); Shoot bends towards light (1); Auxin made at shoot tip (1); Experiment with a covered tip shows no bending (1).	1+1+1+1+1
27	A pair of ovaries produce eggs (1); The egg enters the oviduct (fallopian tube) (1); Fertilisation occurs in the oviduct (1); Zygote moves and is implanted in the uterus (1); Uterus opens into the vagina through the cervix (1).	1+1+1+1+1
	OR: He crossed plants having contrasting traits (1); F1 showed only one parent's trait (1); F2 had both traits in 3 : 1 (1); Traits are controlled by pairs of factors (1); Factors segregate in gametes and are independently inherited (1).	1+1+1+1+1

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 5–8: **Life Processes, Control and Coordination, How do Organisms Reproduce?, Heredity** | Class 10 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(d)** seed	The fertilised ovule becomes the seed.
2	**(a)** 3	The 3 round green seeds are R_yy.
3	**(d)** glomerulus	Glomerulus in Bowman's capsule.
4	**(a)** growth hormone	Pituitary.
5	**(d)** ovaries	Ovaries produce eggs.
6	**(b)** XY	One X and one Y.
7	**(d)** mitochondria	Aerobic respiration in mitochondria.
8	**(b)** auxin	Auxin elongates cells on the shaded side.
9	**(d)** spores	Sporangia release many spores.
10	**(c)** all tall	Tallness is dominant.
11	**(b)** ATP	ATP.
12	**(d)** chemotropism	Chemical stimulus.
13	**(b)** pollination	This is the definition of pollination.
14	**(c)** traits are inherited independently	Independent assortment.
15	**(a)** dialysis	Haemodialysis.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Provides nutrition and oxygen to the embryo (1); Removes waste from the embryo to the mother's blood (1).	1+1
17	A trait gained during the lifetime of an individual (1); Example: loss of body weight due to starvation (1).	1+1
18	CO_2 from air through stomata, water from soil (1); Sunlight from the Sun and chlorophyll in leaves (1).	1+1
19	Electrical impulse changes into a chemical signal (1); Chemical crosses the gap and starts a new impulse (1).	1+1
20	Binary fission: a cell divides into two daughter cells (1); Multiple fission: a cell divides into many daughter cells at once (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	DNA carries the information for building body design (1); Copies of DNA pass the same body design to the offspring (1); Slight inaccuracies in copying create variation (1).	1+1+1
22	Round yellow x wrinkled green gives round yellow F1 (1); F2 shows round green and wrinkled yellow (1); New combinations show independent assortment (1).	1+1+1
23	Alveoli provide a large surface area (1); Thin walls enable gas exchange (1); Rich capillaries take oxygen and give CO_2 (1).	1+1+1
24	Leaves fold when touched (1); Water leaves the cells of the pulvinus (1); Cells shrink and the leaf droops (1).	1+1+1
	OR: Zygote divides to form the embryo in the ovule (1); The ovule develops a tough coat and becomes a seed (1); The ovary grows and ripens to form the fruit; petals, sepals and stamens wither (1).	1+1+1
25	Tall (T) is dominant over dwarf (t) (1); In Tt, tallness is expressed (1); Dwarfness appears only in tt (1).	1+1+1
	OR: Anaerobic respiration in oxygen-poor muscles (1); Lactic acid builds up (1); Cramps relieved by rest or massage (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Mouth: salivary amylase on starch (1); Oesophagus: peristalsis (1); Stomach: HCl, pepsin, mucus (1); Small intestine: bile, pancreatic and intestinal juices (1); Large intestine absorbs water; rectum egestion (1).	1+1+1+1+1

	OR: Receptor -> sensory neuron -> spinal cord (1); Relay neuron -> motor neuron -> effector (1); Reflex: involuntary and fast (1); Voluntary: brain involved and slower (1); Reflex examples: knee jerk (1).	1+1+1+1+1
27	Nucleus elongates and divides (1); The cytoplasm then constricts and splits (1); Two daughter Amoebae are formed (1); Division can occur in any plane (1); Leishmania divides in a definite orientation with respect to the whip-like structure (1).	1+1+1+1+1
	OR: Parents RRYY x rryy (1); F1 all RrYy, round yellow (1); F1 forms four gametes RY, Ry, rY, ry (1); F2 round yellow 9, round green 3, wrinkled yellow 3, wrinkled green 1 (1); Ratio 9 : 3 : 3 : 1 proves independent assortment (1).	1+1+1+1+1

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